

Breast Implant-Associated Anaplastic Large Cell Lymphoma in Australia: A Longitudinal Study of Implant and Other Related Risk Factors

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Disclosures: Prof Deva is an advisor, research coordinator, and educator for Mentor (Johnson & Johnson), Allergan, Sientra. and Motiva (Establishment labs). Prof Magnusson is an advisor and speaker for Allergan and Mentor (Johnson & Johnson). Prof Prince is an advisor and research coordinator for Allergan. The other authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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Level of Evidence: 2 (Risk)

Abstract

Background: Breast implant-associated anaplastic large cell lymphoma (BIA-ALCL) is an emerging cancer that has been linked to the use of textured devices. Recent increase in number and frequency of cases has led to worldwide regulatory action.

Objective: We aimed to longitudinally study the disease in Australia since the index case was first reported in 2007.

Methods: Confirmed historical cases were collected and then prospectively analyzed from October 2015 to May 2019. Clinical and implant exposure data was determined and compared with company sales data for four devices to generate implant-specific risk.

Results: A total of 104 cases of BIA-ALCL were diagnosed in Australia with exposure to 149 unique breast implants. The mean age of patients was 48.2 years (range, 22.4-78.5 years). They had an average time from implantation to diagnosis of 6.8 years. 51.7% of implants utilized in this cohort were Allergan Biocell devices. The indication for implant usage was for primary cosmetic augmentation in 70%, post breast cancer reconstruction in 23% and following weight loss/pregnancy in 7%. The majority of women presented with early (stage 1) disease (87.5%). The risk for developing BIA-ALCL ranged from 1 in 1947 sales (95% CI 1199-3406) for Silimed Polyurethane devices to 1 in 36730 (95% CI 12568-178107) for Siltex imprinted textured devices.

Conclusions: Implants with higher surface area/texture seem to be more associated with BIA-ALCL in Australia. Recent regulatory action to suspend, cancel, or recall some of these higher risk devices is supported by these findings.

The development of a unique T cell anaplastic non-Hodgkin's lymphoma related to breast implants, breast implant-associated anaplastic large cell lymphoma (BIA-ALCL) was first reported in 1997.¹ In recent years, the number and frequency of diagnosed cases appears to be increasing with over 700 confirmed cases² and 33 deaths reported globally.³ The epidemiology of BIA-ALCL shows some unique features with a predilection for Caucasian women.² There are only 3 known cases in women with Asian descent and 1 case in a woman from African descent.² The tumor presents in most cases with an effusion in the space between breast implant and capsule between 7-9 years following insertion of the implant.⁴ It can occur both following breast implant surgery for breast cancer reconstruction or following cosmetic augmentation. Whilst the majority of patients present in early stage disease, some can present with solid tumor invading the peri-implant capsule to subsequently spread to adjacent tissue and draining lymph nodes.⁵ A number of countries have reported their series and confirmed a higher risk for textured breast implants.⁶⁻⁸ Textured implants were introduced as early as the late 1960s in an attempt to provide stability and reduce encapsulation. The first texture to be introduced was Polyurethane (Surgitek) followed by other technologies designed to alter the outer silicone shell.⁹ There are a number of varying surface textures, classified by Jones et al into 4 grades using measurement of surface area and surface roughness.¹⁰ Analysis based on tally of confirmed cases and implant sales have shown that implants with higher surface area and surface roughness confer significantly higher risk to patients.^{6,11} A unifying hypothesis that BIA-ALCL is caused by a combination of texture, contamination by bacteria and stimulation of genetically susceptible T cells over time has been proposed for pathogenesis of this cancer.^{12,13} Other triggers of chronic inflammation including particulates, friction and allergy have also been proposed.²

We have previously published epidemiology of this disease in Australia and New Zealand^{6,11} and now include the most updated case details and implant-specific risk calculations for four distinct types of textured implant.

METHODS

Case Collection

This study was approved by the Macquarie University Ethics Secretariat and complied with guidelines for the conduction of human research. All known cases of BIA-ALCL in Australia have been collected since the index case was diagnosed in 2007. From 2007 to 2015, forensic analysis and self-reporting were utilized to collect data. A partnership between clinicians utilizing breast implants, hematology/oncology, pathology, the Australian Breast Device Registry and Cancer Registries has ensured that since October 1, 2015, prospective notification and data collection of new cases has been undertaken. Demographic, treatment and implant type and exposure times were collected both retrospectively beginning in October 2015 for cases prior and prospectively for cases from this time to May 2019. All patients have confirmed pathology and clinical and pathological staging. Patients were followed up 6 monthly for ongoing surveillance by their treating multi-disciplinary team and any disease progression or recurrence was recorded.

Sales Data

Data on sales of respective textured implants associated with diagnosed cases were obtained on a yearly basis from Allergan, Mentor (Johnson & Johnson), Nagor, and Silimed from 1999 to 2019. For Silimed, the sales of implants did not increase significantly until after 2008, following Australian regulatory approval. These implants were also withdrawn from use in 2015, following regulatory action. For analysis, these data were converted to integral years to allow a discrete time analysis, with analysis restricted to subjects who underwent implantation during or after 1999 within Australia and who had been exposed to a single type of texture only. For comparison, we also performed the risk calculation adding the last implant for patients with multiple implants. We did not include smooth implant exposure in our analysis, as there were no single implant or last implant exposure cases for smooth implants in our cohort. Incidence was calculated using the Poisson distribution and exact confidence intervals. The cumulative proportion of subjects with ALCL exposed to these implants was determined using the Kaplan-Meier estimate, with confidence intervals obtained using a parametric bootstrap. Comparison between groups was performed using discrete-time survival analysis¹⁴ using exact-like logistic regression. All calculations were performed using the R language¹⁵ and packages boot elrm¹⁶

and epitools.¹⁷

Comparison of Surface Area/Roughness to Risk

Previously published data on the surface area/roughness of the 4 available implants (Allergan, Mentor, Nagor, and Silimed)¹⁰ were compared to calculated implant-specific risk.

Clinical Indication and Assignment

Patients were assigned as to indication for breast implant surgery. These included patients following cancer reconstruction, patients who had breast surgery following weight loss/breast ptosis, and patients who underwent primary cosmetic breast augmentation.

For cosmetic breast augmentation, the procedure was classified into those performed by board certified (members of the Royal Australasian College of Surgeons) (C) versus those performed by non-certified (NC) practitioners. Until 2018, with the change of regulation in two States, cosmetic augmentation by NC practitioners was not necessarily performed in a licensed operating facility. Frequency of sales of Nagor, Mentor, and Silimed to C vs. NC practitioners were obtained and compared to the frequency of BIA-ALCL cases in primary cosmetic augmentation cases. Only cases where a single implant led to development of BIA-ALCL were included in this analysis.

RESULTS

A total of 104 patients have been diagnosed with BIA-ALCL since 2007 in Australia. These women have been exposed to a total of 149 implants. The mean age of patients was 48.2 years (range, 22.4-78.5 years). They had an average time from implantation to diagnosis of 6.8 years (range, 0.1-27.0 years). Forty-nine women had BIA-ALCL on the left side (47.1%) whilst 55 (52.9%) had the diagnosis on the right. There were no patients with bilateral disease. Seventy-seven patients (74%) had exposure to a single implant only with the remaining 27 (26%) having exposure to multiple implants. Table 1 details the multi-implant exposure patients and reports the number of implants (range, 2-5) and details the last implant type in these

patients prior to development of BIA-ALCL. None of these patients had smooth implants as the last implant of exposure prior to developing BIA-ALCL. Eleven patients from the reconstructive group had prior tissue expansion before implant placement. Of these nine had subsequent exposure to a single implant whilst the remaining two had multiple implants prior to developing BIA-ALCL.

143 implants were inserted in Australia whilst 6 were inserted overseas (5 Thailand, 1 Peru) (Table 2). A total of 99 of the 104 patients had implantation in Australia and these cases were included in the risk assessment.

Indication for Surgery

A total of 95 patients had recorded indications for surgery and information available on their treating practitioner. The majority of these patients (67) had breast implants for cosmetic augmentation (71%). Twenty-one (22%) had implants for breast cancer reconstruction and the remaining 7 (7%) had implants for weight loss surgery (augmentation mastopexy). All procedures performed for breast cancer reconstruction and weight loss (augmentation mastopexy) were performed by Australian certified surgeons.

For the cosmetic only indication, there were 49 cases accounting for 73% of the total number of cases of BIA-ALCL whose primary procedure was performed by a non-certified practitioner. This compares with only 18 cases (27%) whose primary cosmetic augmentation was performed by an Australian certified surgeon.

Staging, Clinical Outcome, Mortality

Table 3 lists presentation and clinical staging of patients with BIA-ALCL with comparison to previously reported data. The majority of women (87.5%) presented with Stage 1 disease with 57.7% presenting with stage 1a (neg) disease. This group had tumor cells detectable only in the seroma but with no detectable tumor in the capsule or capsular lining.

There was an additional death reported during this last review, bringing the total number of deaths in Australia from BIA-ALCL to 4. Three were related to disease progression and 1 was related to iatrogenic complications of treatment for

BIA-ALCL. The percentage rise in new cases is 43.8% from December 2016 to March 2018 and 57.6% from April 2018 to May 2019.

Implant-Specific Risk

Single Implant Calculation

Collectively implants from the four companies that provided access to sales data (Mentor, Allergan, Nagor, and Silimed) comprised 87% of implants used in patients with BIA-ALCL. Figure 1 summarizes cumulative proportion of BIA-ALCL for these four implant types per 10,000 implants.

Table 4 outlines the implant-specific risk based in patients with single textured implant exposure only. The highest risk was calculated for Silimed Polyurethane implants at 1 case for every 2596 implants sold (95% CI 1486-5024). The odds ratios for developing ALCL were as follows: Biocell vs. Siltex was 6.28 (95% CI 2.30,25.92) $p < 0.0001$; Silimed PU vs. Siltex was 12.46 (95% CI 3.40,80.14), $p < 0.0001$; and Nagor vs. Siltex was 4.12 (95% CI 1.08,19.74), $p < 0.04$.

Last Implant Calculation

Table 5 outlines the implant-specific risk based in patients based on the last implant type. These included patients with both single and multiple implant exposure. The highest risk was calculated for Silimed PU at 1 case for every 1947 implants sold (95% CI 1199-3406).

The odds ratios for developing ALCL were as follows; Biocell vs. Siltex 3.77 (95% CI 1.84,9.10) $p = 0.0001$, Silimed PU vs. Siltex 6.87 (95% CI 2.68, 21.08), $p < 0.0001$ and Nagor vs. Siltex 2.02 (95% CI 0.69, 5.95), $p = 0.2$.

Surface Area/Roughness and Relation to Implant-Specific Risk

Both parameters were fitted separately using the survival model to delineate if there was a relationship between increasing surface area and surface roughness with increasing risk of BIA-ALCL and shown in Table 6.

The effect of roughness was highly significant ($p = 0.0001$) for incidence of BIA-ALCL. The effect of surface area was also similarly highly significant ($p = 0.0007$). These findings confirm that higher surface roughness and area are correlated with

increasing risk of BIA-ALCL for these 4 implant types. Figure 2 shows the increasing level of risk with increasing roughness/surface area for both single and last implant for the 4 implant types studied.

Impact of Certification

We were able to confirm the proportion of implants sold to non-certified vs. certified practitioners for 3 implant types – Nagor, Silimed PU, and Mentor for the period of study. Table 7 summarizes analysis of frequency of BIA-ALCL cases arising from non-certified vs. certified practitioners following primary cosmetic augmentation where a single implant was utilized prior to development of BIA-ALCL. For these three manufacturers, we can conclude that the proportion of BIA-ALCL from non-certified practitioners is significantly higher as compared to the sale of implants to non-certified practitioners ($p = 0.002$ Mentor; $p=0.0004$ Nagor; $p=0.007$ Silimed PU).

DISCUSSION

The latest analysis of risk of BIA-ALCL in the Australian population has confirmed that implants with higher surface area/roughness carry higher risk. The majority (87.5%) of patients are now being diagnosed in early stage through better awareness and improved surveillance. These patients are treated with surgical removal of implant and capsule alone, without the need for adjuvant therapy and have an excellent prognosis.⁴

These updated risk data support the recent regulatory action to suspend or cancel devices that have higher-grade texture in order to reduce the forward incidence of this cancer. The French regulator, the ANSM, was the first to suspend a range of textured devices based on published and independent risk analysis for the French cohort.¹⁸ This was followed by the FDA decision to initiate a recall of the Biocell product, reversing their earlier decision, based on a rise in the number of deaths related to this device.³ Allergan then chose to initiate a voluntary global recall of their Biocell textured implants and tissue expanders.¹⁹ This action is supported by our findings that the risk for Allergan Biocell (Grade 3) device has been steadily increasing since the time of our first publication. The previous cancellation of the Silimed PU device by regulators, albeit for other reasons,⁹ also ensures that the two

highest risk devices are now no longer available for implantation in Australia. The recent decision by the Therapeutic Goods Administration to suspend products that have been associated with cases of BIA-ALCL in Australia was made independently of our risk calculation and with access to unique case load, sales data combined with statistical and internal laboratory analysis.²⁰ The suspension of Grade 3 and 4 devices (Eurosilicone, Polytech Polyurethane) is consistent with our findings with respect to risk of BIA-ALCL in devices with high surface area/roughness. The suspension of the Nagor textured device (Grade 2, open salt loss manufacturing method) is also supported by our findings that the risk and confidence intervals are within the range of other higher risk devices. Their decision for permitting the ongoing use of the Siltex (Grade 2, imprinted texture) device can also be justified based on the lower level risk calculated by comparison to other devices. It is possible, however, that for Grade 2 devices, the signal for BIA-ALCL may be delayed and that over time, the difference between various grades of textured device may normalize and approach a more consistent range. Further analysis of the latency before developing BIA-ALCL for varying grades of implant are required to verify this hypothesis and to ensure the safety of other textured devices with respect to the risk of BIA-ALCL.

We chose to report two methods of implant risk calculation in this study. We have previously utilized single implant exposure only in our risk calculation,^{6,11} conceding that this method most likely underestimates the true risk of BIA-ALCL when using sales data as it does not reflect the actual type and number of implants that are implanted in patients. We have previously outlined the limitations of this method¹¹ including underreporting and/or errors in reporting of cases and the use of sales data as a substitute for actual implantation numbers. The use of sales data to derive denominators, however, is a valuable proxy to the true risk. The use of last implant of exposure includes patients who develop BIA-ALCL after multi-implant exposure into the risk calculation and does yield a higher risk as compared with the single implant calculation. The difficulty with using last implant is that the relative weighting or risk for patients with multiple implant exposure is ascribed to a single implant type and therefore may over-represent one device as compared to others. What was consistent, however, using both methods, is the relative rates risk of each of the devices studied remains the same, with the Grade 4 surface imparting the

highest risk. We have previously discussed the relevance of this finding with respect to the unifying hypothesis of BIA-ALCL.¹¹ Whilst our initial analysis suggested a higher frequency of *Ralstonia pickettii* in BIA-ALCL specimens,²¹ further microbiome analysis from prospective sample collection is ongoing and will further elucidate the frequency of these and other bacteria utilizing fresh tumor samples. Prospective clinical data collated by maturing breast implants registries will be key in delineating the true incidence and implant-specific risk. The International Collaboration of Clinical Quality Breast implant Registries (ICOBRA) aims to harmonize and combine collective implant outcomes and will, in time, provide a better estimation of forward risk.²²

The confirmation that increasing surface area and surface roughness are significantly associated with risk of BIA-ALCL across these four varying textured devices is consistent with previous published data.^{6,11,23,24}

For cases arising out of primary cosmetic augmentation, subset analysis of the impact of certification has yielded an unexpected finding, with a higher frequency of observed cases for practitioners who are non-certified by our national body as specialist surgeons. The observation of clusters of multiple cases of BIA-ALCL cases arising from single surgeons has hinted at the potential impact of perioperative and intraoperative factors.¹¹ Additionally, many of these cases were, until recently, performed in unlicensed facilities, potentially compounding the risk by failure to regulate consistent standards of safety and infection control. Further study is required to investigate the relative impact of certification/training and licensing of facilities on the risk of developing BIA-ALCL. It is timely that we look to ensure that the highest standards of clinical practice (implant choice, informed educated consent, shared decision making, cooling off period prior to elective breast implant surgery, bacterial mitigation²⁵ and ongoing surveillance post implantation) are outlined and enforced so as to promote patient safety. Attention should also be focused on ensuring appropriate training, practice and certification are attained before clinicians are permitted access to these Class 3 medical devices. These findings echo experience with pelvic and vaginal mesh, where lack of practitioner education, training and certification have been associated with a higher rate of adverse events.^{26,27}

CONCLUSIONS

The risk of BIA-ALCL in Australia is rising and is significantly more associated with breast implants of higher surface area/roughness. The levels of risk reported for Australia justifies recent worldwide regulatory action. The training, experience and certification of the practitioner using these devices and the facilities where these procedures are performed may impact on the risk of BIA-ALCL and requires further study. The use of sales data analysis in Australia has provided a clearer picture of the emerging patterns of this disease and valuable risk vs. benefit information to guide clinicians, regulators and patients to ensure ongoing safety and efficacy.

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[AQ: Please review the URLs in References #16 and 27 for accuracy.]

Table 1. Number of Patients with BIA-ALCL with Single vs. Multiple Implant Exposure

	No. of patients	Percentage
Implant exposure		
Single	77	74%
Multiple	27	26%
No. of implants		
2	17	16.3%
3	5	4.8%
4	2	1.9%
5	3	2.0%
Last implant in patients with multiple implants		
Allergan Biocell	14	51.9%
Silimed Polyurethane	6	22.2%
Mentor Siltex	5	18.5%
Nagor	1	3.7%
Polytech Polyurethane	1	3.7%

Table 2. Type and Number of Implants Utilized in the 2019 Patient Cohort and Comparison with Previous Years

Manufacturer	Surface	Grade ¹⁰	2016	2018	2019	Percentage
Silimed	Polyurethane	4	12	20	28	18.8%
Polytech	Polyurethane	4	0	1	2	1.3%
Surgitek	Polyurethane	4	1	1	1	0.7%
Allergan/Inamed	Biocell (Salt loss)	3	26	51	77	51.7%
Eurosilicone	Salt Loss	3	0	0	1	0.7%
Nagor	Nagotex (Salt Loss)	3	4	6	8	5.4%
Mentor	Siltex	2	2	6	17	11.4%
PIP	Textured	2	0	4	6	4.0%
Mentor	Smooth	1	2	2	2	1.3%
Unknown	Smooth	1	2	2	5	3.4%
Unknown	Texture	Unknown	0	0	2	1.3%
Total			50	93	149	

Table 3. TNM Staging and Deaths for Patients Over 3-Year Time Period with Percentage Increase in Number of Cases

TNM	Stage	2016	2018	2019	Percentage	Deaths
T1N0M0	1A(neg)	28	38	60	57.7%	0
T1N0M0	1A(pos)	7	15	25	24.0%	0
T3N0M0	1C	6	6	6	5.8%	0
T4N0M0	2A	3	5	8	7.7%	3
T4N1M0	3	1	1	1	1.0%	1
T4N2M0	3	1	1	3	2.9%	0
T4N2M1	4	0	0	1	1.0%	0
Percent increase in cases			43.8	57.6		

Table 4. Risk of BIA-ALCL for 4 Implant Types Based on Sales Data and Single Implant Exposure. Mean and 95% Confidence Intervals Shown.

Implant	Mean no. of implant sales per single case of BIA-ALCL (95% CI)	Rate per 10,000 implant years (95% CI)
Silimed PU 95% CI	2596 (1486-5024)	0.59 (0.30-1.02)
Biocell 95% CI	3194 (2387-4379)	0.38 (0.27-0.50)
Nagor 95% CI	6024 (2768-16417)	0.22 (0.08-0.47)
Siltex 95% CI	36730 (12568-178107)	0.050 (0.010-0.147)

Table 5. Risk of BIA-ALCL for 4 Implant Types Based on Sales Data and Last Implant Exposure. Mean and 95% Confidence Intervals Shown.

Implant	Mean no. of implant sales per single case of BIA-ALCL (95% CI)	Rate per 10,000 implant years (95% CI)
Silimed PU 95% CI	1947 (1199-3406)	0.78 (0.45-1.27)
Biocell 95% CI	2436 (1888-3200)	0.49 (0.37-0.64)
Nagor 95% CI	5164 (2506-12844)	0.25 (0.10-0.52)
Siltex 95% CI	15741 (7640-39153)	0.117 (0.047-0.241)

Table 6. Relationship Between Surface Area/Risk and Risk of BIA-ALCL

		Odds ratio (95% CI)	p value
Surface roughness	Per 10 units	1.071 (1.036,1.104)	0.0001
Surface area	Per 10 units	1.201 (1.085,1.318)	0.0007

Table 7. Percentage of BIA-ALCL Cases Generated by Non-Certified Practitioners with Confidence Intervals Compared with Sale Proportion of Implants to Non-Certified Practitioners

	Proportion non-certified (95% CI)	Sale proportion	p value
Mentor	80.0% (28.4%, 99.5%)	15.0%	0.002
Nagor	83.3% (35.9%, 99.6%)	15.0%	0.0004
Silimed	91.7% (61.5%, 99.8%)	52.0%	0.007

Figure Legend

Figure 1. Kaplan Meier plots for each of the 4 implants outlining cumulative proportion of patients with ALCL per 10,000 implants.

Figure 2. Risk of BIA-ALCL per 10,000 implant years versus various implant types. The relationship between surface area/roughness is significant.

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Figure 1

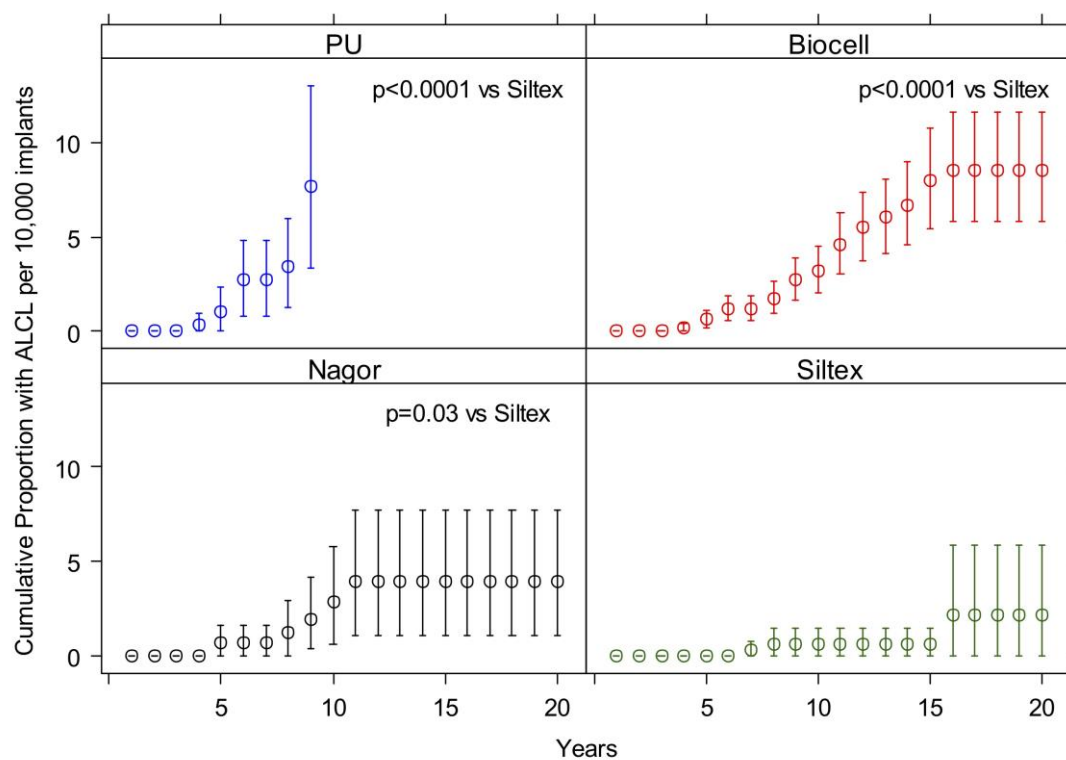
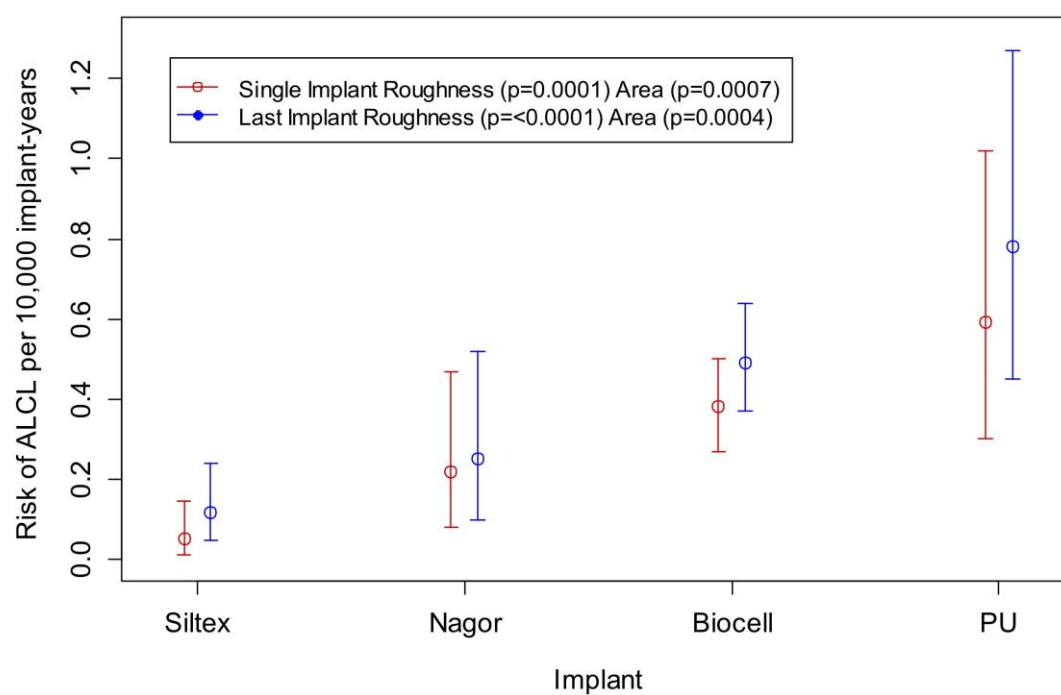


Figure 2



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